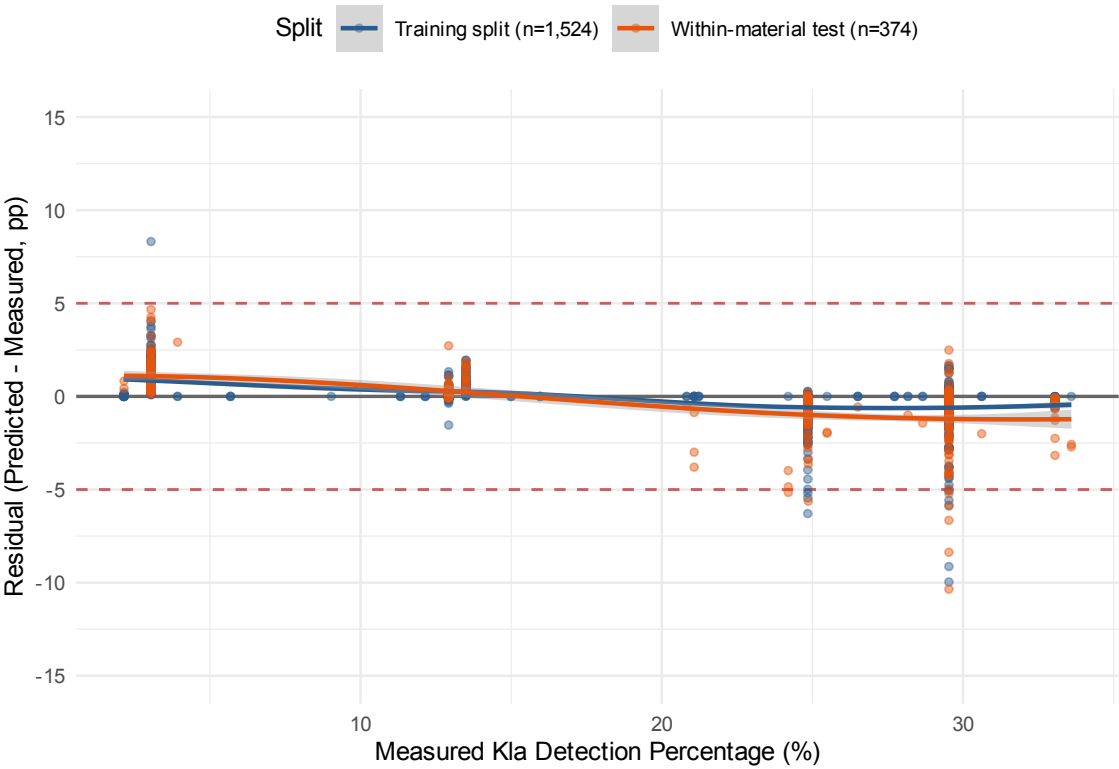


Figure 4: Model Residual Diagnostics on Training and Test Splits

Evaluation of prediction errors across the range of Kla percentages and error normality

A. Residuals vs True Measured Kla %

Dotted red lines denote ± 5 percentage point tolerance band; curves show LOESS fit



B. Error Density Distribution

Sharp peak around 0 indicates unbiased predictions without variance inflation

